Object Oriented Programming in java | classes, object New keyword | JAVA FULL COURSE #12.

Object Oriented Programming →

```
String name = "Aditya";
int age = 28;
int rollno = 101;
String college = "IIT Guwahati";
String name2 = "Rohit";
int age2 = 28;
int rollno2 = 102;
String college2 = "IIT Guwahati";
Void print (String name, int age, int rollno, String college) {
  System.out.Println ( ____ );
}.
```

Problems → • Need a lot of independent variable to represent one student.

• If I pass the student to a ~~pr~~ function i need to pass all 4 variables.

• I need to create new set of variables for new student

• No authority over data.

Object Oriented Programming → Real word problem to mimic

name
age
rollno
College

⟹

Nam	age
roll no	College

{ name, rollno, age, college }.

Real world object

Arjun, varun.

Programming world

```
class student {
  String name;
  int age;
  int rollno;
  String college;
}
```

→ Blue print of an object

Human
- student
- → son/daughter
- → employ.

Data Type
- → prem
- → Non pre
- → Arrays.
- → objects
- → Userdefine data types

userdefine.

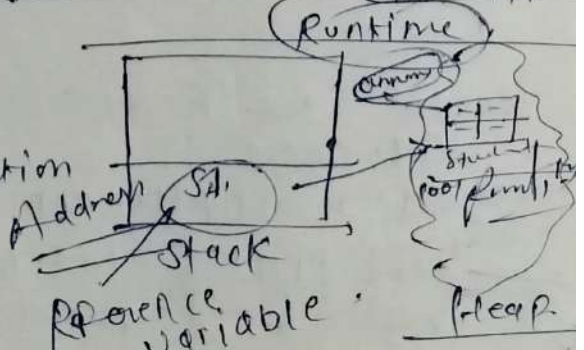
object
Runtime
↓ error
↓ dynamic

Compile time
Redline error

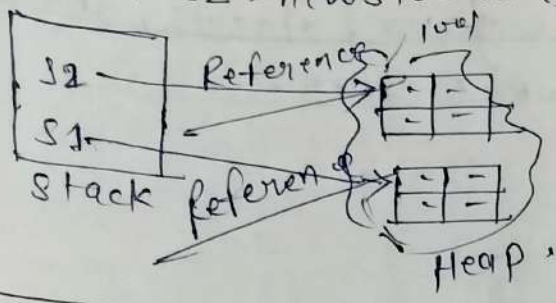
```
Student s1;
s1 = new student ();
```

```
Student s1 = new Student ();
s1.name = "Aditya";
s1.age = 28
s1.rollno = 101
s1.college = IIT Guwahati
```

→ Dynamic Memory location



Student s2 = newstudent();



s2.name = "Rohit";
s2.rollno = "age";
s2.age = "28";
s2.college = "IIT GUWAHATI";

```
class student {
  String name;
  int age;
  int rollno;
  String college;

  student s1. new student ();
  student s2. new student ();

  s1.name = "Aditya"
  s1. age = 28;
  s1. rollno = 101;
  s1. college = "IIT GUWHATI"

  s2. name = "Rohit"
  s2. age = 28;
  s2. rollno = 102;
  s2. college = "IIT GUWHATI"
```

Print System.Out.Println (s1.nam);

Naming Convention

Good practice

Variables | Premitive / reference
↳ Camel case

firstName, lastName,

class → ~~String with~~ Starting with Capital letters
student, Human, Char

functions → Camel case

Student
↳ character →
↳ Behaviour →

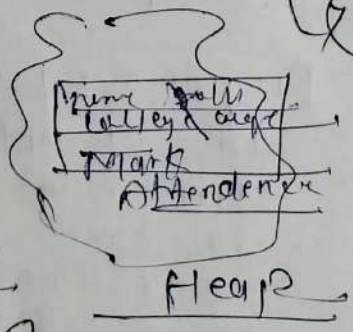
class student()

using → variables

Animal → characteristic
↳ Type → Number of legs, Number of Color, white, fur)
↳ Behaviour
↳ (make sound, eat, fur)
↳ function
↳ does Something

```
class Animal () {
  String type;
  int no of legs;
  void make sound () {

}
```




```
class Student {
  String name;
  String Colleg;
  int rollno;
  int college;
  void marksAttendence () {

}
```

character

→ Behaviour

```
Student s1 = new student ();
  s1. name ;
  s1. markAttendence ();
  Void print (student s) {
  System.Out.Print (s.name, s.age)
```

Real world object	→	Digital object
Animal	→	✓
Student	→	✓
Bank account	→	✓

```
Public class Demo {
Public static void main (string [] args) {
 student s1 = new student ();
 student s2 = new student ();

 s1. name = "Sachin";
     s1. age = 20;
     s1. college = "IES BHOPAL";
     s1. rollno = 95;

     s2. name = "Raj";
     s2. age = 20;
     s2. college = "IES Bhopal";
     s2. roll no = 55;
}}
```

output

Attence marked by Sachin
Attence marked by Raj
Sachin 2095 IESBh
Raj 2095 IESBhopal

```
s1.
~~void mark~~ Attendence. () {
    s2. Attendence. () {
    s1. print ();
    s2. print ();
```

```
class student {
 String name;
 int age;
 int rollno;
 String college;
 void markAttendence. () {
  System. Out. Println ("~~Aditya Sachi~~
  ~~Marked~~ Attendence. Marked by" + name);
}
 Void print () {
  System. Out. Println (name + "," + age + "," rollNo + "," +
                        college);
}}
```